Ranks and Moments of (Hyper)elliptic Curve Families

Yuval Amit[1] Lucas Chen[2] Joshua Im[3]
Devayani Pradhan[4]
Mentor: Steven J. Miller

SMALL REU, Williams College

[1]yuval.amit33@gmail.com, [2]lucasch@uchicago.edu,[3]jri@tamu.edu,
[4]pradhand@umich.edu

Yale REU Conference
July 21, 2026



What is an Elliptic Curve?

We can see an elliptic curve as a particular **plane curve**: the solution set of the equation:

$$E/K : y^2 = x^3 + Ax + B, \quad A, B \in K$$

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The points (x, y) that satisfy an elliptic curve E lie in K^2, where K can be any field.

Elliptic Curves - Example

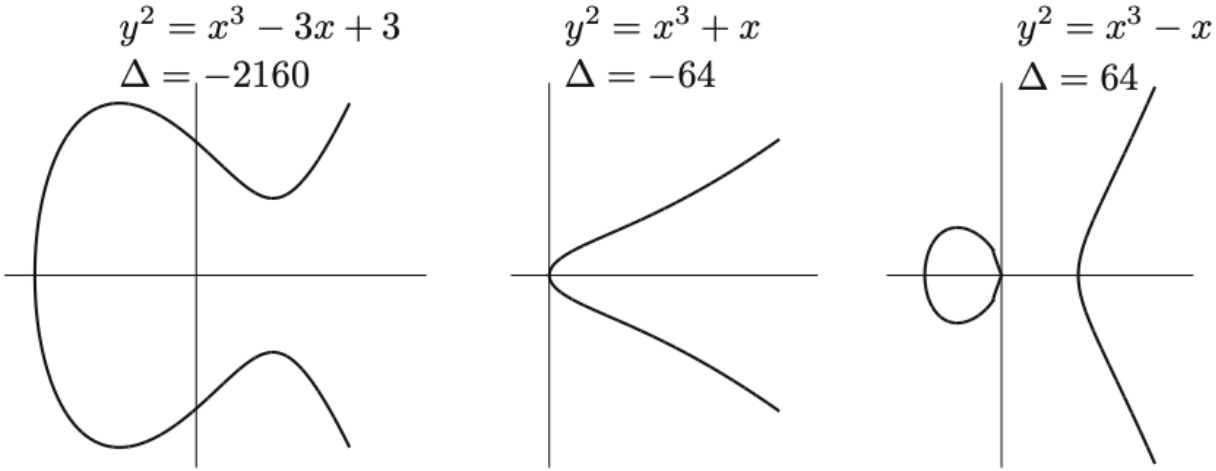


Figure: Three elliptic curves, for $K = \mathbb{R}$. (Silverman 2009)

The Group Law

Any elliptic curve turns out to also be an **abelian group**.

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Definition

Let $P, Q \in E(K)$ be points on the curve. Define R to be the third point that the line passing through P and Q passes. Define $P + Q$ to be the reflection of R w.r.t the y-axis. Define

$$P + Q + R = O := \mathrm{id}_E.$$

Then $(E(K), +)$ forms an abelian group.

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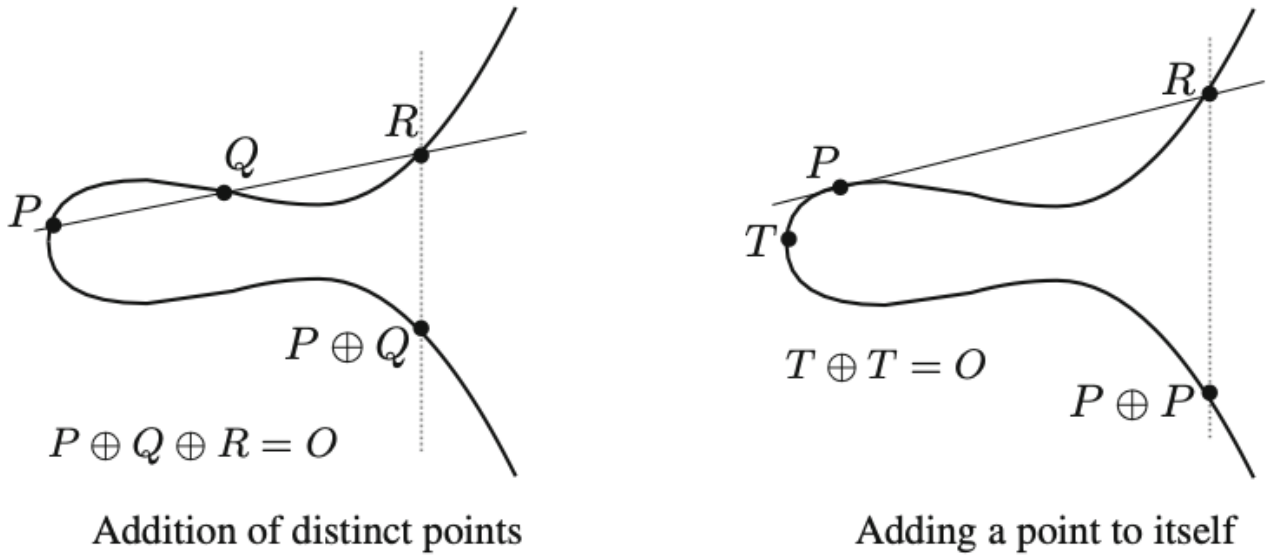


Figure: The composition law. (Silverman 2009)

The Mordell-Weil group

A K-**rational point** is a point $(x, y) \in K^2$ on E/K.

- For a field K with closure $\overline{K}$, the set of K-rational points $E(K)$ is a subgroup of $E(\overline{K})$.

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The theorem below is a fundamental result in the field.

Theorem (Mordell-Weil, 1929)

The set of K-rational points $E(K)$ is a finitely generated abelian group. That is,

$$E(K) \cong \mathbb{Z}^r \times \mathbb{Z}/p_1^{k_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_r^{k_r}\mathbb{Z},$$

where each p_i is prime.

The number of free generators r is called the **rank** of E.

Hyperelliptic Curves

We begin by defining a k-group scheme as a k-scheme with a canonical morphism $m : G \times_k G \to G$. The group scheme satisfies commutative diagrams equivalent to the standard group axioms. Now to generalize the definition of an elliptic curve, we note that it has a generically étale morphism of degree 2 onto a genus 0 curve, so we extend this property to curves of higher genus than 1: this gives us the standard definition of a hyperelliptic curve. A hyperelliptic curve, due to the increase in genus, does not have a rigid identification of valid degree 0 line bundles with compatible triples of closed points on the curve. In order to rectify this problem and establish geometric content on the natural group of degree 0 line bundles, i.e. the Picard zero group, we establish the Picard functor as a representable functor for the group scheme-Hopf algebra dual category via Grothendieck's definition. Because we would like to be able to take group quotients we also must establish the functor on a finer topological site than the Zariski topology, for which we employ the big-fppf flat site as per Grothendieck. We finally define the relative Jacobian variety for a smooth proper relative curve with geometrically connected fibers as $\mathrm{Pic}^0_{C/S}$, into which the original curve may embed as a closed immersion via the Abel map, reliant on a nonvanishing condition. We transition into French at this point for maximal faithfulness to the original formalism. Supposons k algébriquement clos. Pour tout k-schéma localement algébrique $\mathcal{X}$, nous identifons $\mathcal{X}(k)$ á l'ensemble des points ferm'es de $\mathcal{X}$. On sait que $A \mapsto A \cap \mathcal{X}(k)$ est une bijection de l'ensemble des parties fermées (resp. ouvertes, resp localement fermées, resp. irréductibles, resp. constructibles) de $\mathcal{X}$ sur l'ensemble analogue pour $\mathcal{X}$. Soit $\mathfrak{f}$: $\mathbf{Z}_k \to \mathfrak{G}$ l'homomorphisme tel que $\mathfrak{f}(1) = x$. Soit $\mathfrak{H}$ l'image fermée de $\mathfrak{f}$ (I, §2, 4.10) ; d'après I, §2, 4.15, $\mathfrak{H} \otimes_k \bar{k}$ est l'image fermée de $\mathfrak{f} \otimes_k \bar{k}$; elle est portée par l'adhérence E de l'image de $|\mathfrak{f} \otimes_k \bar{k}|$. Mais $E \cap \mathfrak{G}(\bar{k})$ est l'adhérence de

Hyperelliptic Curves (for real)

A **hyperelliptic curve** of genus g is a natural generalization of the elliptic curve. For a polynomial $f(x) \in K[x]$, it is cut out by the equation

$$\mathcal{X}/K : y^2 = f(x), \qquad \deg f(x) = 2g+1 \text{ or } 2g+2$$

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Any algebraic curve C generates a **Jacobian variety** denoted $J(C)$, a geometric object which is an abelian group. For an elliptic curve E, $J(E) \simeq E$.

The Jacobian satisfies Mordell-Weil and has an equivalent notion of rank.

Reduction to $\mathbb{F}_p$

The locus and point count of $\mathbb{F}_p$-valued points of E is closely related to the rational points of E.

We consider a specific map for the $\mathbb{F}_q$ case, the **Frobenius endomorphism**.

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Theorem (Freshman's dream)

For a point $(x, y) \in \mathbb{F}_p^2$, and a polynomial p where $p(x, y) = 0$, we have

$$p(x^p, y^p) = \left[p(x, y)\right]^p = 0.$$

We can distinguish the rational $\mathbb{F}_p$-valued points from others in the algebraic closure $\overline{\mathbb{F}}_p$ as the fixed points of the Frobenius endomorphism.

Frobenius Trace

Suppose we have an (hyper)elliptic curve $\mathcal{X} : y^2 = f(x)$ defined over $\mathbb{F}_p$. Then, define

$$\#\mathcal{X}(\mathbb{F}_p) = \#\{(x, y) \in \mathbb{F}_p^2 \mid y^2 \equiv f(x) \pmod{p}\}$$

Theorem (Hasse, 1933)

In the case of elliptic curves E, $|(p+1) - \#E(\mathbb{F}_p)| \leq 2\sqrt{p}$.

Definition (Frobenius Trace)

We call $a_{\mathcal{X}}(p) = (p+1) - \#\mathcal{X}(\mathbb{F}_p)$ the **Frobenius trace** of $\mathcal{X}$ at a prime p.

(This is legitimately the trace of a Frobenius operator on a very specific module!)

Moments of (Hyper)Elliptic Curves

Our research focuses on (hyper)elliptic curves $\mathcal{X}$ over $\mathbb{Q}(T)$. Then the elliptic curve obtained by the specialization $T = t$ is well-defined over $\mathbb{Q}$, which we call $\mathcal{X}_t/\mathbb{Q}$.

Definition (rth Moment)

Fix a prime p. Let $a_{\mathcal{X}_t}(p)$ be the Frobenius trace of $\mathcal{X}_t$ at p. The **rth moment** of $\mathcal{X}$ is defined by

$$A_{\mathcal{X},r}(p) = \frac{1}{p}\sum_{t=0}^{p-1} a_{\mathcal{X}_t}(p)^r.$$

Nagao's Conjecture

Nagao conjectured on the relationships between the first moment and the rank.

Conjecture (Nagao)

For an elliptic curve $E/\mathbb{Q}(T)$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{p\leq N} -A_{E,1}(p)\log p = \text{rank}(E/\mathbb{Q}(T)).$$

Conjecture (Nagao, Generalized)

For a hyperelliptic curve $\mathcal{X}/\mathbb{Q}(T)$ such that the Jacobian variety of $\mathcal{X}$ has no subvariety over $\mathbb{Q}$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{p\leq N} -A_{\mathcal{X},1}(p)\log p = \text{rank}\, J_{\mathcal{X}}(\mathbb{Q}(T)).$$

Nagao - Why do we care?

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Known results on existence of (hyper)elliptic curves over $\mathbb{Q}(T)$:

- For any g, there exists infinitely many hyperelliptic curves of rank 0, 1, 2.
- Rank could be arbitrarily large.

In general, it is not known if there exists an hyperelliptic curve for a specific (g, r).

Main Results

We state our main results.

Theorem

Assume generalized Nagao's conjecture. Then there exists an hyperelliptic curve $\mathcal{X}/\mathbb{Q}(T) : y^2 = f(x, T)$ *of genus* g *with* $\operatorname{rank} J_{\mathcal{X}}(\mathbb{Q}(T)) = r$ *if*

- $5 \leq r \leq 2g - 5$ *if* $(g, r) \neq (6, 6), (9, 6), (12, 6)$, *and* $(15, 6)$, *with the assumption of Goldbach's conjecture,*
- $g < r \leq 2g + 1$,
- $2g + 1 \leq r \leq 4g + 2$.

So, with the assumption of Goldbach and generalized Nagao, we can find any hyperelliptic curve with a fixed genus and rank, in the given range.

Constructions

1. $5 \leq r \leq 2g-5$:

$$y^2 = x(x^{q_1} - 1^{q_1}) \cdots (x^{q_n} - n^{q_n})$$

(where q_is are pairwise coprime)

2. $g < r \leq 2g+1$:

$$y^2 = x(x^2+1) \cdots (x^2+m^2) \cdot (x^2-1^2) \cdots (x^2-n^2)$$

3. $2g+1 \leq r \leq 4g+2$:

$$y^2 = x(x^2+a_1^4) \cdots (x^2+a_m^4) \cdot (x-b_1^2) \cdots (x-b_{2n}^2)$$

Choosing appropriate coefficients, we can force a specific rank.
Proof uses Galois theory, algebraic geometry, and complex analysis.

Normalized Frobenius

Recall: For a prime p and E_t a family of elliptic curves:

- The Frobenius trace $a_p(t) = a_{E_t}(p)$
- The normalized Frobenius trace: $f_p(t) = \frac{a_p(t)}{\sqrt{p}}$
- Gives information on number of roots of E_t over F_p

Question:

- How does f_p grow as $p \to \infty$?
- Can we apply standard signal reconstruction algorithms to f_p?

Fourier Ratio

We define the ℓ_1 norm and ℓ_2 norm of any $f : \mathbb{F}_p \to \mathbb{C}$ by

$$\|f\|_1 := \sum_{x \in \mathbb{F}_p} |f(x)| \qquad \|f\|_2 := \left(\sum_{x \in \mathbb{F}_p} |f(x)|^2 \right)^{1/2}$$

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$$FR(f) = \frac{\|\hat{f}\|_1}{\|\hat{f}\|_2}$$

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The magnitude of $FR(f)$ tells us how much information is needed to recover f. We q apply this to elliptic curves by studying $FR(f_p)$.

Main Result - Elliptic Curves

Theorem

Let $E : y^2 = x^3 + A(T)x + B(T)$ be a one-parameter family of elliptic curves over $\mathbb{Q}$. Define $f(t) = a_p(E_t)/\sqrt{p}$ for $t \in \mathbb{F}_p$. Then, the Fourier ratio of f satisfies

$$FR(f) = \frac{\|\widehat{f}\|_1}{\|\widehat{f}\|_2} \asymp p^{1/2}.$$

Proof

Compute explicitly with Plancherel: $||\hat{f}||_2 \asymp \sqrt{p}$ since

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- Cauchy Schwarz: $||\hat{f}||_1{}^2 \leq p||\hat{f}||_2{}^2$ so $||\hat{f}||_1 \ll p$.

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- Then, $||\hat{f}||_1 \geq \dfrac{||\hat{f}||_2^2}{||\hat{f}||_\infty} \gg p$ since Michel 1995 yields

$$||\hat{f}||_\infty = \max_{m \in \mathbb{F}_p} \frac{1}{p^{1/2}} \sum_{x \in \mathbb{F}_p} f(x)\chi(-mx) = O(1)$$

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Thus, $||\hat{f}||_1 \asymp p$.

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Theorem

Let $E_T : y^2 = x^3 + A(T)x + B(T)$ *be a one-parameter family of* **hyperelliptic** *curves over* $\mathbb{Q}$. *Define* $f(t) = a_p(E_t)/\sqrt{p}$ *for* $t \in \mathbb{F}_p$. *Then, the Fourier ratio of* f *satisfies*

$$FR(f) = \frac{\|\widehat{f}\|_1}{\|\widehat{f}\|_2} \asymp p^{1/2}.$$

Generalization to Hyperelliptic Curves

Key Result: For elliptic curves, Michel 1995 says

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Idea: Extend to isotrivial families and then to hyperelliptic curves.

- Apply Lefschetz-Grothendieck trace formula:

$$\sum_{x \in \mathbb{F}_p} f(x)\chi(-mx) = \sum_{i=0}^{2} (-1)^i \mathrm{tr}(\mathrm{Frob}|_{H^i})$$

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- Show $H^0, H^2 = 0$.
- Show eigenvalues of H^1 are bounded by $\sqrt{p}$.

References



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